Continuing Declines in NSW Birds

Which species are still declining, and where and which kinds

Client briefing · CS-13 Combined Breeding Analysis team · 18 October 2026

Data: NSW Bird Atlas (1977–2020) and DES backlog (to 2025)

THE QUESTIONS

Q1 **Which species, if any, are continuing to decline?**

27 native species fell in every period from 1977 to 2020, led by Hooded Robin, Pallid Cuckoo, Little Eagle; 23 more show a new decline.

Q2A **Are declines concentrated in particular regions?**

Declines occur in every zone; the typical species lost 12% of its squares on the coast and 1% inland.

Q2B **Are declines concentrated in particular habitats or ecological groups?**

Insect-eating woodland birds are over-represented among decliners: 18 of the 27 continuing decliners eat mainly insects.

HOW TO READ THIS REPORT

This report is a companion to 'Understanding Bird Breeding in NSW' and 'Hidden Pressures'. Each page asks one research question, answers it with a chart, and explains what it means for protecting species. Because survey lists got shorter over time, comparisons are made like-for-like and against the typical species, not on raw totals.

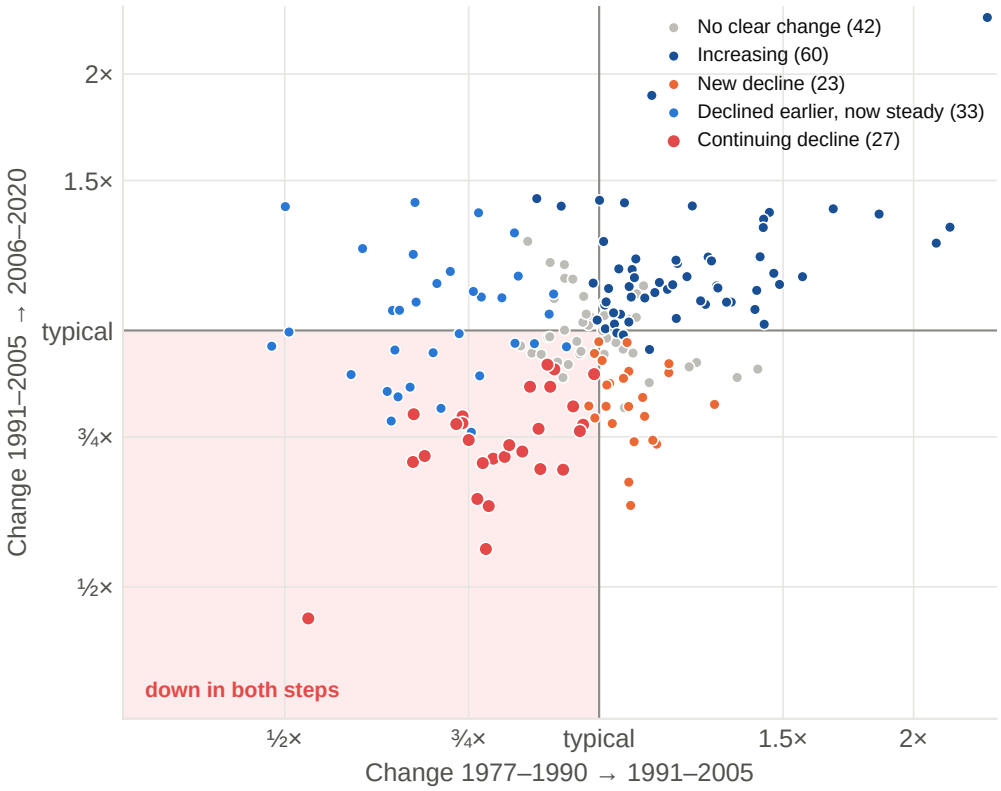
RESEARCH QUESTION 1

Which species, if any, are continuing to decline?

Why this matters: A species that fell once and then levelled off needs different action from one that is still falling. A decline that continues across three periods means whatever is driving it has not stopped.

27 native species declined in each period from 1977 to 2020

Change vs typical species: 1977–1990 → 1991–2005 (across) and 1991–2005 → 2006–2020 (up)



Still declining (change 1977→2020)

Hooded Robin	-76%
Pallid Cuckoo	-56%
Little Eagle	-53%
Buff-rumped Thornbill	-51%
Scarlet Robin	-51%
Little Lorikeet	-51%
Dusky Woodswallow	-46%
Restless Flycatcher	-45%
Crested Shrike-tit	-44%
Horsfield's Bronze-Cuckoo	-43%
Jacky Winter	-43%
Rainbow Bee-eater	-42%
Striated Thornbill	-42%
Yellow-billed Spoonbill	-41%
White-naped Honeyeater	-39%
Mistletoebird	-39%
Fuscous Honeyeater	-38%
Brown-headed Honeyeater	-36%
Brown Treecreeper	-32%
Yellow Thornbill	-26%
Dollarbird	-25%
Southern Whiteface	-24%
Gang-gang Cockatoo	-22%
White-throated Treecreeper	-22%
Yellow-rumped Thornbill	-18%
Peaceful Dove	-18%
Yellow-faced Honeyeater	-11%

ANSWER

- Of 185 widespread native species, 27 show a continuing decline: down from 1977–1990 to 1991–2005 and down again to 2006–2020, with both the recent and overall drops clear.
- The steepest are Hooded Robin, Pallid Cuckoo, Little Eagle, Buff-rumped Thornbill, Scarlet Robin. 18 of the 27 mainly eat insects, such as robins, thornbills, treecreepers and woodswallows.
- Breeding points the same way: for 12 of the 20 continuing decliners with enough breeding records, a smaller share of sightings showed breeding than for the typical species.
- A further 23 species show a new decline since the 1990s (for example Chestnut-rumped Thornbill, White-browed Babbler, Leaden Flycatcher), while 33 fell earlier but have been steady since. 60 are increasing.

What it means: The continuing decliners are the clearest candidates for priority action and for checking against threatened-species listings. The 'new decline' group is an early warning list worth watching closely.

Caution: The Atlas ends in 2020, so 'continuing' means through 2006–2020. Introduced European Goldfinch, Common Starling also declined steadily and are excluded from the list. Results describe where species were recorded, not how many birds there are.

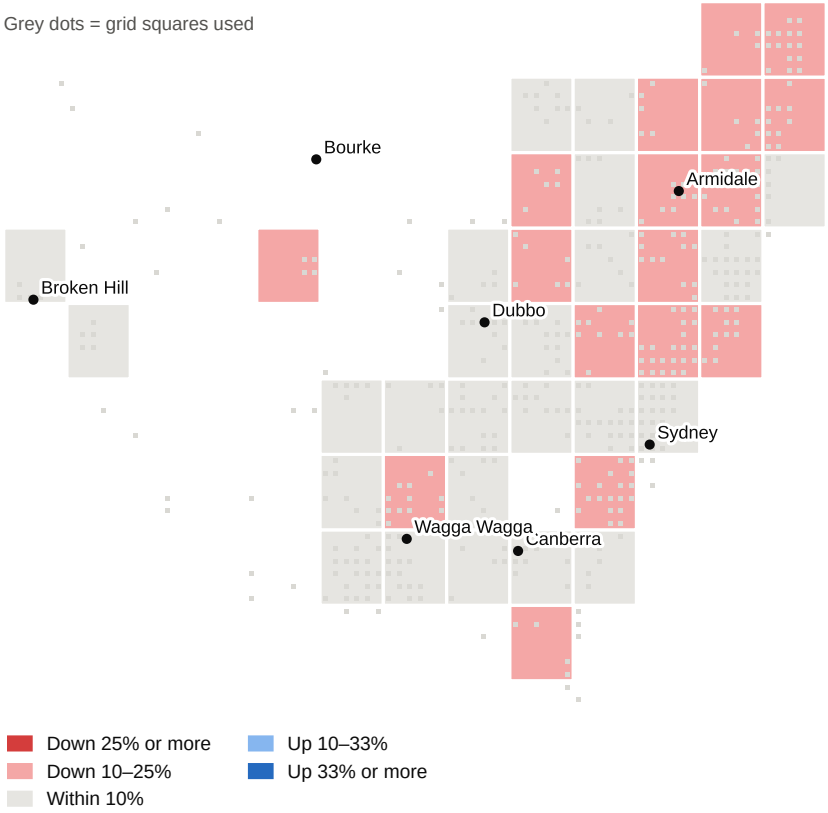
RESEARCH QUESTION 2A

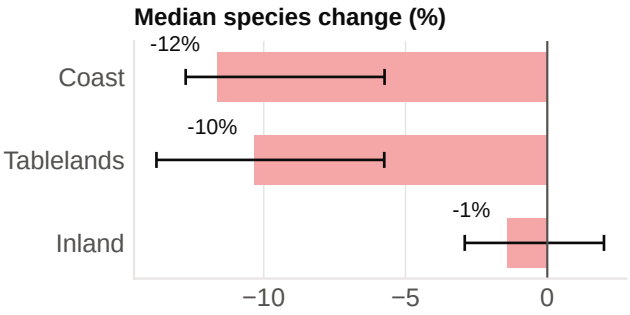
Are declines concentrated in particular regions?

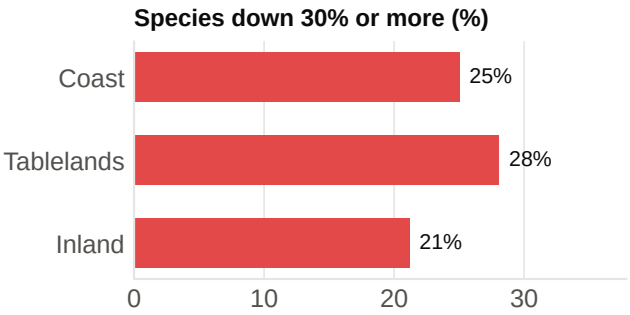
Why this matters: Declines clustered in one part of NSW point to local land use or management; declines spread everywhere point to a statewide pressure. The answer decides whether action should be regional or statewide.

Declines are widespread, a little steeper on the coast than inland

Median species change in share of squares where recorded, 1977–1990 → 2006–2020 (fair comparison)







ANSWER

- Declines are found in every zone, but are mild inland. The typical species lost 12% of its squares on the coast, 10% on the tablelands and 1% inland.
- 28% of tablelands species, 25% of coastal species and 21% of inland species lost at least 30% of their squares.
- On the map, 41% of 41 cells show the median species down by 10% or more; cells that held steady are scattered rather than forming one block.
- The continuing decliners are not tied to one zone: of the 27, 11 fall hardest on the coast, 9 fall hardest inland, 7 fall hardest on the tablelands.

What it means: Presence declines are broad rather than local, so statewide measures (such as woodland protection and restoration rules) matter more than any single region. The first report found breeding fading most inland; together, the two suggest that inland birds may still be present but breeding less, while on the tablelands and coast some species are being recorded in fewer places altogether.

Caution: Few far-western squares were surveyed in all three windows, so the inland result rests mostly on the central west. Region names are still to be confirmed with the client, so zones are used here.

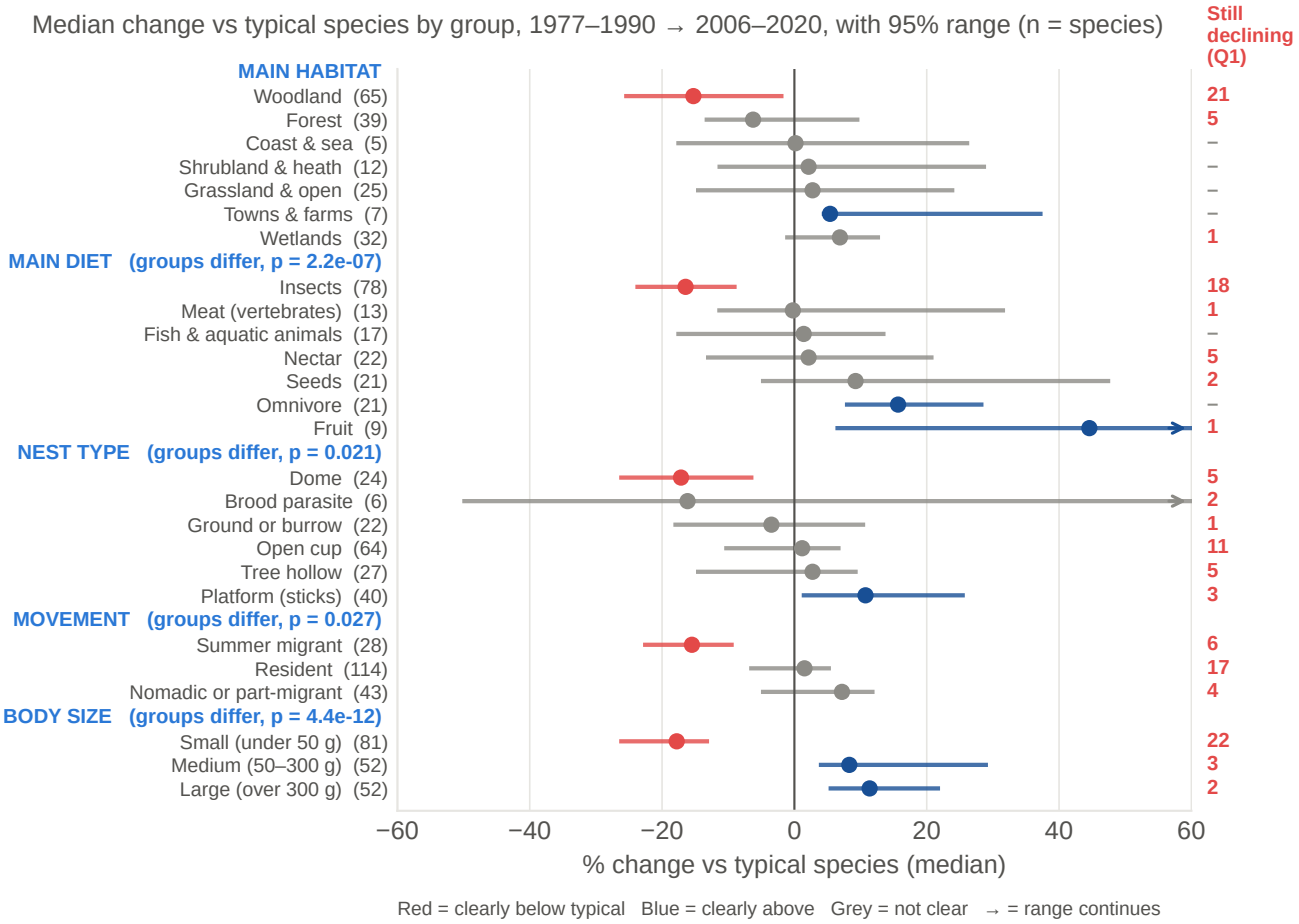
RESEARCH QUESTION 2B

Are declines concentrated in particular habitats or ecological groups?

Why this matters: Species that share a habitat, diet or nesting need probably share a pressure. Finding that shared trait means one action can help many species at once.

Declines are concentrated in small, insect-eating woodland birds

Median change vs typical species by group, 1977–1990 → 2006–2020, with 95% range (n = species)



ANSWER

- Yes. Groups clearly below the typical species are: small birds under 50 g (-18%), dome-nest builders (-17%), insect-eaters (-16%), summer migrants (-15%), woodland birds (-15%).
- Insect-eaters stand out: 18 of the 27 continuing decliners eat mainly insects, and the typical insect-eater changed by -16% relative to the typical species.
- By habitat, woodland birds fared worst (-15%); groups doing clearly better than typical include fruit-eaters (+45%), omnivores (+16%), large birds over 300 g (+11%), stick-platform nesters (+11%).
- The groups differ by more than chance for body size, main diet, nest type, movement (after allowing for testing five traits); for main habitat the difference is suggestive only.

What it means: The shared profile of a declining bird is a small, insect-eating woodland species. Keeping and restoring woodland with a native ground layer, fallen timber and shrubs — the places these birds find insects — is the action most likely to help many decliners together.

Caution: Traits are provisional (compiled for this analysis) and should be checked against the team's trait table. Traits overlap — most woodland birds also eat insects — so groups are not independent.

SUMMARY

What this means for protecting bird species

Still declining	27 native species fell in every period from 1977 to 2020, led by Hooded Robin, Pallid Cuckoo, Little Eagle; 23 more show a new decline.
Where	Declines occur in every zone; the typical species lost 12% of its squares on the coast and 1% inland.
Which kinds	Insect-eating woodland birds are over-represented among decliners: 18 of the 27 continuing decliners eat mainly insects.

SUGGESTED NEXT STEPS

1. Check the continuing decliners against NSW and national threatened-species listings.
2. Add the 'new decline' species to an early-warning watchlist and repeat this check as new data arrive.
3. Prioritise woodland ground-layer, shrub and fallen-timber restoration, which supports insect-eating birds.
4. Confirm the provisional trait table against the team's trait file before sharing the group results.
5. Use DES surveys from 2021 onwards to test whether the continuing declines carry on after 2020.

ABOUT THE DATA AND METHOD

- Records: 6,619,049 usable species records in NSW. Trends use Atlas surveys only, 1977–2020; DES recording differs, so its rates are not pooled with the Atlas. eBird is excluded, as agreed with the client.
- Fair comparison: single-month lists with 15–45 species, 4 lists per grid square per window in the 432 squares surveyed in all three windows, averaged over 25 draws.
- Each species is compared with the typical species; uncertainty ranges come from 400 resamples of squares.
- Traits are provisional, compiled for this analysis using the team's categories.
- All settings can be changed in the accompanying notebook; tables for every chart are in the outputs folder.